

MongoDB set-up and installation

MongoDB comes in both 32-bit and 64-bit versions for different OS platforms. You can get the installation details from the MongoDB site.



Note: It is always preferable to install the 64-bit version for better performance.

Installation on Ubuntu

Installation on Ubuntu is very easy unlike the complexity with a Windows machine. If you want a simple installation, then follow the steps mentioned below.

Run the following commands from a terminal:

```
$ sudo apt-get update
$ sudo apt-get install mongodb-org
```

Start the *mongod* service by executing the commands shown below:

```
$ sudo service mongod start
```

Open another terminal window and execute the following command:

```
$ mongo
```

The above command opens the shell prompt, which will connect to MongoDB.

Your MongoDB installation is now complete and ready to be used.

Sometimes, the above method may not work for Debian Linux. Don't be disheartened; download the .gz file of MongoDB. For me, it is *mongodb-linux-x86_64-debian71-3.0.1.gz*. Extract it by using the following command:

```
# tar -xvzf mongodb-linux-x86_64-debian71-3.0.1.gz
```

After that, run the command shown below:

```
export PATH=$PATH:/mongodb/mongodb-linux-x86_64-debian71-3.0.1/bin
```

The above command adds the MongoDB binary directory to the Linux path.

Run the following commands to run the service:

```
# cd /mongodb/mongodb-linux-x86_64-debian71-3.0.1/bin
# ./mongod --dbpath=/data/mongodb3
```

A better way to do the above task is to put both commands in one file saved as *mongo1.sh*, then run the *mongo1.sh* shell script.

For the Mongo terminal, run the following commands:

```
# cd /mongodb/mongodb-linux-x86_64-debian71-3.0.1/bin
# ./mongo
```

You can save both commands in a file saved as *mongo2.sh*, and then run the *mongo2.sh* shell script.

You can install MongoDB in Windows too. Check the steps of the installation under the NoSQL tab of *L4wisdom.com*.

MongoDB with Python

Now let's study how Python and MongoDB interact. You can run MongoDB and Python programs on Linux. In order to install the *pymongo* module, you can take advantage of the PIP program. Use the following command:

```
# pip install pymongo
```

After entering the command, the progress of the installation will be displayed.

Now let's explore some Python code for MongoDB.

Insert data with Python. MongoDB stores the data in the JSON format. I am presenting a JSON file named *test.json* for testing:

```
{
  "Records": [
    {
      "record": "from socrates to sartre: the philosophic quest (paperback)",
      "ID": 39
    },
    {
      "record": "the magic of believing by bristol claudie m.-english-embassy books-paperback (paperback)",
      "ID": 40
    },
    {
      "record": "haunted: hardy boys (paperback)",
      "ID": 42
    },
    {
      "record": "the intelligent investor (paperback)",
      "ID": 44
    },
    {
      "record": "understanding & worshipping sri chakra (paperback)",
      "ID": 45
    },
    {
      "record": "pmr: i too had a love story (paperback)",
      "ID": 55
    }
  ]
}
```